

CHARLES MICHAEL F. CAYNO

Pasig City, Metro Manila, Philippines | (+63) 968 2671 491 | cfayno@gmail.com
LinkedIn: <https://www.linkedin.com/in/charlescayno/>

PROFESSIONAL SUMMARY

Results-driven Senior Power Supply Design Engineer with over 6 years of professional experience in the design, development, simulation, and validation of high-efficiency AC-DC and DC-DC power supply systems. Recognized expert in switch-mode power supply (SMPS) topologies, custom magnetic component design, high dv/dt & thermal stress analysis, and EMI/EMC mitigation. Proven capability in digital power control, automated multi-instrument testing across 20+ power IC products with Python, global project execution (including US engineering assignment), conference paper authoring (PCIM), and product qualification under global regulatory standards. Skilled in leading engineering projects, cross-functional collaboration, and mentoring technical teams.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Power Electronics & Topologies:** AC-DC Converters, DC-DC Converters, Flyback, Buck, Boost, Flyboost, Resonant Converters (LLC), Switch-Mode Power Supplies (SMPS), Power Factor Correction (PFC), Switch Valley-Fill (SVF), Zero-Voltage Switching (ZVS), GaN / PowiGaN Technology, SiC (Silicon Carbide), 1700V High-Voltage Systems, Qi Wireless Charging, USB Power Delivery (USB PD 3.0 / 3.1 EPR).
- **Circuit Design & Simulation:** SIMetrix, SIMPLIS, LTspice, MATLAB, Simulink, Schematic Capture, Transformer & Inductor Design (PQ, EFD, EE Cores), Ferrite Core Selection, Winding Design.
- **PCB Layout & Tools:** Altium Designer 26, OrCAD Allegro, KiCad, High-Voltage Layout Design, Creepage & Clearance Compliance, Thermal Layout Optimization.
- **Compliance & Testing:** EMI/EMC Compliance (CISPR 22 / EN 55022 Class B), Conducted & Radiated Immunity Testing (ESD, Surge), High dv/dt Testing Hardware, Extreme High-Temperature Testing (125°C Tj).
- **Test Automation & Software:** Python 3 Automation Libraries, GUI Development, Automated Multi-Instrument Testing Across 20+ Power IC Products, Test Automation (GPIB, VISA, Instrument Control), C/C++, MATLAB.
- **Embedded & Digital Power Control:** Microcontrollers (MCUs), Digital Signal Processors (DSPs), Bluetooth Low Energy (BLE) Protocols, Nordic, Microchip.

PROFESSIONAL EXPERIENCE

Power Integrations Netherlands B.V. (Philippine Branch) — Pasig City, Metro Manila

Senior Power Supply Design Engineer | April 2024 – Present

- Lead the end-to-end design, topology selection, and hardware development of high-efficiency AC-DC and DC-DC power supply systems for smart home automation, high-voltage industrial, server auxiliary, and smart lighting applications.
- Architect high-power-density power supply designs featuring GaN / PowiGaN, SiC (Silicon Carbide), and 1700V InnoMux2 technologies with optimized thermal management and low EMI.
- Execute detailed magnetic design (PQ, EFD, EE cores), thermal dissipation calculations, and PCB layout reviews to comply with international safety, high dv/dt, and EMC standards.
- Execute specialized US engineering assignments (May – June 2026) collaborating with US lead engineering teams on high-temperature SilentBridge power platforms.
- Author technical papers, collaborate with cross-functional engineering teams, and mentor junior engineers in power electronics design and testing.
- **High-Voltage Server Auxiliary Power Supply (215W / 1700V Integrated Switcher Architecture) [Ongoing Design]:** Architecting and prototyping a 215W peak (170W continuous) 48V/3.55A auxiliary power supply for data center and server power delivery architectures operating from a 700V–860V DC bus using a 1700V integrated power switcher IC. Designing custom PQ32/20 power magnetics, executing custom secondary IC trim configuration, developing specialized

interposer test hardware with 3D-printed forced-air thermal mounting, and building automated Python ATE test routines.

- **DER-1113 (80W Flyback Converter using TopSwitchGaN & PQ 35/35 Core):** Designed an 80W power supply achieving 91.78% peak efficiency at 230 VAC using custom PQ 35/35 litz wire magnetics. Solved TopSwitchGaN IC overvoltage damage caused by TVS breakdown voltage mismatch. Optimized radiated EMI using Material 43 ferrite suppression; delivered functional marketing samples for global customer evaluation.
- **DER-1081 (50W Emergency Smart Lighting Supply):** Developed a slim-profile (9.78 in x 1.06 in x 0.63 in) 50W flyback power supply with 1 CV and 1 CC output, high power factor (>0.904), low THD (<9.09%), and 1-wire PWM dimming. Resolved complex startup faults and VDS voltage stress issues; delivered validated Rev B & C marketing samples to key stakeholders.
- **0195b — SilentBridge High-Temp Testing & RCA (US Business Trip):** Stationed in the US (May – June 2026) collaborating with lead engineers (Thomas & Jared) on the SilentBridge high-power platform. Conducted extreme high-temperature stress testing (125°C Tj, 95°C ambient) and root-cause failure analysis (RCA) on shorted LV/HV FETs.
- **InnoMux2-1700V & Inno3-SiC High dv/dt Testing Hardware:** Evaluated 1700V InnoMux2 max power capabilities for utility grid and industrial applications. Designed specialized high dv/dt test hardware for Silicon Carbide (InnoSwitch3-SiC F package) and evaluated automotive InnoSwitch3-AQ dv/dt characteristics during DOPL.
- **DER-1050 (82W Smart Clothes-Dryer Rack Flyback Supply):** Designed an 82W flyback converter with 2 CV and 1 CC output using InnoMux2-EP and Switch Valley-Fill (SVF) passive PFC. Featured at PCIM 2025.

Power Integrations Netherlands B.V. (Philippine Branch) — Pasig City, Metro Manila

Power Supply Design Engineer | March 2022 – April 2024

- Designed high-power-density DC-DC power converters and LED drivers; performed root-cause failure analysis on field returns and prototype units.
- Evaluated early-stage power IC products, identifying corner-case limitations and operational boundaries in coordination with IC design teams.
- Designed custom high-frequency transformers and inductors tailored to target electrical and thermal requirements.
- Developed embedded C firmware for microcontroller-based digital power supply controls and communication protocols.
- **DER-1024 (65W Dual USB-C Power Supply):** Designed a 65W dual USB-C shared capacity power supply with current sharing using InnoSwitch3-Pro PowiGaN and Injoinic USB PD controllers. Showcased design at APEC 2024.
- **DER-1015 (140W USB PD 3.1 / EPR Power Supply):** Provided technical leadership for a 140W (280W peak power) USB PD 3.1 supply using InnoSwitch5-Pro and HiperPFS-5, optimizing conducted EMI and synchronous rectification ZVS operation.
- **DER-867 (2-Wire BLE Smart Wall Switch):** Developed a smart wall switch using LinkSwitch-TNZ with an automated relay Zero-Voltage Switching (ZVS) timing calibration algorithm to eliminate inrush current. Integrated BLE protocol firmware.
- **DER-727 (75W 2-Stage Dimmable LED Ballast):** Optimized dimming circuitry using LYTSwitch-6 PowiGaN and HiperPFS-5, enabling smooth <1% dimming without shimmer.
- **DER-999 (High-Power Wireless Charger):** Led hardware/software development for a high-power wireless charger utilizing HiperLCS-2; implemented Foreign Object Detection (FOD), Qi wireless standard compliance, and automated telemetry.

Power Integrations Netherlands B.V. (Philippine Branch) — Pasig City, Metro Manila

Applications Evaluation Engineer | January 2020 – March 2022

- Executed fabrication-assembly transfer evaluations, hardware validation, and automated multi-instrument testing across 20+ power IC products (spanning InnoSwitch, TinySwitch, LinkSwitch, and LYTSwitch families).
- Developed Python 3 automation scripts and custom GUI applications for GPIB/VISA instrument control (AC sources, electronic loads, oscilloscopes, power meters), improving test efficiency and data accuracy.

- Evaluated marketing sample boards to fulfill urgent customer requests, verifying compliance with IEC 61000 conducted immunity, surge, and ESD standards.
- Performed detailed teardown analyses of competitor power adapters for internal benchmarking.
- Provided technical field support to regional Field Application Engineers (FAEs) across India, China, and Taiwan; facilitated PSU 101 technical training workshops.
- **DER-916 (65W Dual USB-C Power Supply):** Evaluated dual-port USB PD 3.0 power supply utilizing IP2726 controller; verified compliance with CISPR 22 Class B / EN 55022 Class B EMI standards.
- **DER-945 (75W Dimmable LED Driver):** Designed and tested a 75W 3-in-1 dimmable LED driver (100–300 VAC input, 24V–46V output) using LYTSwitch-8 in Flyboost topology; presented at FAEC 2022.

Power Integrations Netherlands B.V. (Philippine Branch) — Pasig City, Metro Manila

Applications Engineering Intern (Low Power Team) | June 2018 – July 2018

- Supported low-power power supply evaluation projects through data collection, circuit troubleshooting, and EMI mitigation.
- Designed, built, and tested an offline flyback power supply capstone project using TinySwitch technology, including OrCAD schematic capture, PCB layout, transformer winding, and bench testing.

EDUCATION

University of the Philippines Diliman — Quezon City, Metro Manila

Bachelor of Science in Electronics and Communications Engineering | August 2014 – June 2019

- **Capstone Project:** Design and Implementation of an Integrated Motor Drive and Battery Charger with a Bridgeless Boost PFC Converter.
- **Extracurricular:** Shell Eco-Marathon Asia Participant (2018 & 2019).

LICENSES & CERTIFICATIONS

- **Licensed Electronics Engineer** — Philippine Professional Regulation Commission (PRC), Nov 2019
- **Licensed Electronics Technician** — Philippine Professional Regulation Commission (PRC), Nov 2019

TECHNICAL TRAININGS

- Safety Guidelines for High Voltage Systems
- Product Safety Basic Concepts & General Requirements
- Electrical Safety Isolation
- Safety Lab Protocols and Equipment Familiarization
- DOPL and FPGA Training